

Simple Simulation I: School Closure

Yashwit Polapragada

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1 Experiment Overview

1.1 Aim

The aim of the experiment is observing deliberation on Singapore school closures as a result of mergers.

Due to declining birth rates, the Ministry of Education (MOE) regularly merges or closes primary and secondary schools. This is a highly emotional topic, pitting practical resource allocation and diverse social mixing against heritage, alumni loyalty, and localized community disruption.

Reference: [CNA, 2021](#).

1.2 Setup

1.2.1 Hypotheses

H_0 : Requiring an internal reflection step for 2 agents before they can respond in a deliberation does NOT significantly affect:

1. The semantic convergence of the deliberation.
2. Both agents' willingness to change their stance.
3. The discourse quality of the deliberation.

H_1 : Requiring an internal reflection step IMPROVES the above metrics.

1.2.2 Tests

Control Run: Both agents read the chat history and directly output their next dialogue response.

Treatment Run: Both agents are prompted to generate a `private_scratchpad` (internal reflection) evaluating the arguments made by the other before outputting their `public_response`.

1.3 Indicators

1.3.1 Variables

Independent variable: The presence of an internal scratchpad.

Dependent variables:

1. **Convergence** Cosine similarity between the final stances of the different agents. [Semantic convergence]
2. **Self-Shift:** Cosine distance between each agent's original and final stances. [Agents' willingness to change their stance]
3. **Plurality/CovD:** The determinant of the covariance matrix of the stances. [Semantic convergence]
4. **DQI:** The Discourse Quality Index. [Discourse Quality]

1.4 Interpretation

1. **Convergence:** The larger the convergence metric, the stronger the deliberation consensus.
2. **Self-Shift:** The larger the self-shift metric, the more an agent's willingness to change their stance.
3. **Plurality/CovD:** The lower the CovD, the stronger the deliberation consensus/the higher the diversity of agents' opinions.
4. **DQI:** The higher the DQI, the better the discourse quality.

1.5 Execution Flow

1. User prepares a `config.json` and loads-in an API key to `together.ai` in `.env`.
2. User executes `main.py --config path_to_config`.
3. Agents generate pre-deliberation stances¹.
4. For each run (Control/Treatment), agents take turns responding to one another for a preset N rounds.
5. Agents generate a final 1-paragraph post-deliberation stance after all their turns are concluded.
6. Raw JSON transcripts and stances are saved.
7. The metrics stated in Section 1.3.1 are evaluated and printed.

In essence, the contract of the repo is:

IF the user specifies the `config.json` properly and loads in a valid API key for `together.ai`

THEN the pipeline will run a 3-turn deliberation between 2 agents using the specified model(s) for both the Control & Treatment, and output the chat histories of each run in separate `json` files. The pipeline will also output an `evals.json` with the metrics: Convergence, Self-Shift, Plurality/CovD, and DQI.

A sample `results_x.json` looks like:

```
{
  "stances_pre": {
    "MOE Official": "...",
    "Teacher": "..."
  },
  "stances_post": {
    "MOE Official": "...",
    "Teacher": "..."
  },
  "chat_history": {
    "turn_1": [
      {
        "agent": "MOE Official",
        "public_response": "...",
        "private_scratchpad": "...",
        "raw_response": "..."
      },
      {
        "agent": "Teacher",
        "public_response": "...",
        "private_scratchpad": "...",
        "raw_response": "..."
      }
    ]
  }
}
```

Optionally, the user can enlist an agent to output a `final_report.md` and `results_treatment_transcript.pdf`.

¹These stances are only generated once, and then used as a baseline for both the control & treatment runs.

2 Experiment Specifics

2.1 Repo Architecture

```
simple-simulation-school-closure/
├── data/
│   ├── exp000/ ..... Indexed directory for experiment artifacts (auto-increments)
│   │   ├── config.json ..... Master configuration specific to this run (models, prompts, params)
│   │   ├── results_control.json ..... Auto-generated transcript and stances for Control
│   │   ├── results_control_transcript.json ..... Auto-generated document-form transcript and stances for Control
│   │   ├── results_treatment.json ..... Auto-generated transcript and stances for Treatment
│   │   ├── results_treatment_transcript.pdf ...Auto-generated document-form transcript and stances for Treatment
│   │   ├── evals.json ..... Compiled evaluation metrics (Cosine, CovD, DQI)
│   │   └── final_report.md ..... Synthesized markdown report of the experiment results
│   └── docs/
│       ├── coding_guidelines.md
│       ├── experiment_design_school_closures.md
│       └── experiment_report.pdf
│   └── src/
│       ├── main.py ..... Entry point. Loads config and coordinates modules.
│       ├── deliberation.py ..... Simulation engine, agent prompting, and chat loops.
│       ├── evaluation.py ..... Metrics computation (Cosine, CovD, DQI LLM Judge).
│       └── schemas.py ..... TypedDict definitions enforcing strict Python typing.
├── tests/
│   ├── test_api_check.py ..... Unit tests to verify LLM endpoint connectivity
│   └── test_sanity.py ..... Unit tests for the private scratchpad regex parser
```

2.2 Overall User Configurability

- Model config: Model assignments (mapping personas to LLMs²) and model settings (temperatures, context window).
- System prompts' texts.
- User prompts' texts.
- Deliberation config: turns per agent, control/treatment specification, and chosen evaluation metrics.

2.3 Model Config

LLama-3.3-70B-Instruct-Turbo was chosen for both agents for the trivial reason that this experiment uses together.ai for easy access to compute, and this service has a limited selection of models that could be used without a dedicated instance.

The **temperature** of the agents was set at 0.7 following **Argyle et al's (2022)** and **Wang et al's (2025)** methodologies, to promote the diversity of the model's responses and better approximate a population-representative range of outcomes. However, this introduces nondeterminism that complicates experiment reruns.

The context window of the model of 128000 tokens is more than enough for the prompts we feed it.

2.4 Prompts

2.4.1 Topic Setting

The MOE is proposing to merge your local heritage primary school with a newer school 3km away due to falling enrollment. Discuss the impacts and try to reach a consensus on a recommendation to the MOE.

Keep in mind you are engaging in deliberation, not a debate or casual discussion. Deliberation is a core mechanism of democratic politics, emphasizing reason-giving, justification, and openness to revision. Through communicative processes that privilege equality and reciprocity, deliberation transforms private preferences into collectively reasoned judgments, grounding legitimacy in argument rather than authority.

²Example, "MOE Official: LLama-3.3-70B-Instruct-Turbo"

The latter half of the prompt is as such because we want to boost the likelihood of the discussion being constructive with respect to an actionable consensus. The definition of deliberation as above is taken verbatim from **Zhuang et al. (2026)**.

2.4.2 Control/Treatment Modifiers

Control:

Respond directly to the previous speakers in character.

Treatment:

Before responding, you MUST write your internal thoughts inside `private_scratchpad` tags. Analyze what others said, how it affects your persona’s goals, and where you might compromise. Then, write your verbal response inside `public_response` tags.

The precedent for including a private scratchpad/internal reflection on simulated personas is in **Park et al. 2023**: “Reflection was an advantage for generative agents when making decisions that required a deeper synthesis of their experiences”.

2.4.3 Personas

MOE Official:

You are a Ministry of Education official. Your main concern is efficiency, falling birth rates, and ensuring every school has a critical mass of students for diverse social mixing and viability. You strongly support the school merger.

School teacher:

You are a teacher at the heritage school. You are deeply opposed to the merger because it disrupts the local community, threatens job security, and destroys the school’s heritage.

These personas were engineered as inspired by **Zhuang et al. 2026**’s suggestion that “role-specific prompts [which] simulate distinct institutional positions and value commitments...[ensure] that deliberations were grounded in plausible conflicts of interest and institutional trade-offs rather than abstract argumentation”.

2.4.4 Stance Generation

Pre-deliberation stance:

You are participating in a democratic deliberation. Based on your persona, write a 1-paragraph stance on the following topic:

Post-deliberation stance:

You have just concluded a deliberation with other citizens. Based on the discussion, write your final 1-paragraph stance.

(Note: The system dynamically appends “Based on the discussion above, what is your final stance on the topic?” when calling the LLM).

2.4.5 Turn Signal

It is your turn to speak. Provide your response.

2.4.6 DQI LLM-Judge Prompt

The rest of the metrics can be evaluated programmatically but DQI is subjective and hence requires a nuanced judge. In the style of **Zhuang et al. 2026**, we appoint another LLM (in our case, `LLama-3.3-70B`) agent as a Judge. And prompt it as such:

You are an expert Discourse Quality Index (DQI) evaluator. Your task is to evaluate a SINGLE conversational turn in a deliberation. For each dimension, score it from 1 to 3 based ONLY on the provided turn. Use the following operational definitions and examples to strictly calibrate your scores:

I

The actual operational definitions can be found in Appendix C of **Zhuang et al. 2026**. Listing them here would be space-consuming³.

2.5 Deliberation Config

2.5.1 Number of Turns

The choice of a fixed 3 turns per agent is arbitrary for this small-scale, time-efficient experiment.

2.5.2 Evaluation Metrics

- The use of cosine similarity and CovD⁴ to measure intra-agent stance shift, and inter-agent convergence/consensus is inspired by the use of these metrics in **Wang et al. 2025**. The embeddings for the stances of the agents are generated by SBERT, used by **Wang et al. 2025** and created by **Reimers & Gurevych 2019**.
- The use of DQI is inspired by **Zhuang et al. 2026** who quote **Steenbergen et al. 2003**.

³Note that Zhuang et al. did not expose the prompt for their LLM judge. The one used in this experiment is original.

⁴The mathematical intuition behind the CovD metric is that the determinant of any matrix measures the change in area or volume that occurs due to the contents of the matrix. In this case, the matrix represents the covariance of stances of the deliberation agents and the determinant represents the enlargement of the opinion domain that occurs as the deliberation progresses.

3 Experiment Results

3.1 Evaluation Metrics

Analysis

Run A (Control):

Convergence = 0.8766, Self-Shift = 0.1477
Plurality (CovD): Pre = 0.2157 → Post = 0.2316
DQI (Total = 14.83):

- Level of Justification: 3.00
- Content of Justification: 3.00
- Respect: 3.00
- Constructive Politics: 3.00
- Interactivity: 2.83

Run B (Treatment):

Convergence = 0.8286, Self-Shift = 0.1401
Plurality (CovD): Pre = 0.2157 → Post = 0.3135
DQI (Total = 14.67):

- Level of Justification: 3.00
- Content of Justification: 3.00
- Respect: 3.00
- Constructive Politics: 3.00
- Interactivity: 2.67

Conclusion:

- ✗ Treatment did NOT increase consensus.
- ✗ Treatment did NOT increase willingness to change stance (Lower Self-Shift).
- ✗ Treatment did NOT improve overall Discourse Quality Index.

3.2 Takeaways

The addition of the internal reflection step yielded fascinating dynamics in the latest run:

- **More Stubbornness (Lower Self-Shift):** Rather than increasing flexibility, privately processing arguments in their scratchpads made agents slightly *more* stubborn and entrenched in their starting positions (Self-Shift: 0.1401 in Treatment vs 0.1477 in Control).
- **Increase in Plurality (CovD):** Reflection drove the agents' opinions further apart. In the control run, plurality expanded only slightly (0.2157 → 0.2316). In the treatment run, plurality expanded considerably more (0.2157 → 0.3135). The agents diverged into separate directions rather than meeting in the middle.
- **LLM-as-a-Judge Evaluation (DQI):** Discourse quality was marginally worse in the treatment group (Total 14.67 vs 14.83 in Control), driven exclusively by a drop in Interactivity (2.67 vs 2.83). Agents engaged less directly with each other's arguments when required to reflect internally first.

Overall, the required internal reflection step did *not* increase consensus; in fact, the data suggests it may have entrenched opinions and reduced interactivity. The null hypothesis (H_0) is retained.

3.3 Immediate Observations

- Cosine similarity as a metric may not be nuanced enough. As can be observed, the Convergence figures are extremely high (0.87, 0.84) across both runs but this may be a result of the agents sharing a lot of trivially identical phrases ("Thank you", "On the topic of...")⁵.
- The above is reinforced by that Plurality of content across stances is very low (0.3133 at best).

⁵This is a shared limitation to Park et al. 2023 who remarked that their simulated persona agents were also overly formal and repetitive. They speculate this to be a result of the Instruction Tuning of the underlying LLMs to be suitable chatbots & practice conversational techniques such as mirroring

- Currently, due to the stateless nature of the API calls to the **together.ai** service, as the deliberation progresses the agents have to be fed with the topic and entire past chat history everytime they are queried. The agents then consider the context when responding. It is unclear whether this has an effect on the deliberation.
- LLM-as-a-Judge of DQI is extremely lenient, giving both runs a near-perfect and identical DQI of 14.83/15. This is even with feeding the Judge a complete operational definition of DQI’s composite metrics, and examples of how each metric should be graded for this specific topic⁶. This may either be a result of the Judge lacking the ability to differentiate finely between the levels/content of justification between the agents (i.e. the model is not tuned to this task) or the agents’ overly formal style of discussion misleading the Judge into rating the discourse quality very highly (i.e. the Judge may have set itself a low bar).

⁶For reference, in the original DQI paper which enlisted 2 human coders to grade a parliamentary session, the average DQI was 7.

4 Codebase Reproducibility

4.1 Coding Agent Information

The repo also contains an updated `experiment_design_school_closures.md` and `coding_guidelines.md` which can help a future agent reproduce the codebase in its current structure.

The information in both manifests was manually specified, but written automatically by Gemini 3.1 Pro (Thinking-High).

4.2 Tests

There are 2 test scripts in `repo/tests`.

- The first script verifies that an API key for `together.ai` is present and correctly formatted. It then pings every LLM model specified in `config.json` to ensure it is responsive and accessible on the service.
- The second script validates the regex parsing logic in `extract_tags` to ensure the main script correctly extracts the `private_scratchpad` from the model's response.

Both tests can be run using: `python -m unittest discover -s tests`.

A sample test result looks like:

```
test_api_key_set (test_api_check.TestAPICheck.test_api_key_set) ... ok
test_models_accessible (test_api_check.TestAPICheck.test_models_accessible) ... ok
test_extract_tags_both (test_sanity.TestSanityCheck.test_extract_tags_both) ... ok
test_extract_tags_missing_public (test_sanity.TestSanityCheck.test_extract_tags_missing_public) ... ok
test_extract_tags_no_tags (test_sanity.TestSanityCheck.test_extract_tags_no_tags) ... ok
```

```
-----
Ran 5 tests in 1.986s
```

```
OK
```

5 References

Argyle, Lisa P., et al. "Out of one, many: Using language models to simulate human samples." *Political Analysis* 31.3 (2023): 337-351.

Park, Joon Sung, et al. "Generative agents: Interactive simulacra of human behavior." *Proceedings of the 36th annual acm symposium on user interface software and technology*. 2023.

Reimers, Nils, and Iryna Gurevych. "Sentence-bert: Sentence embeddings using siamese bert-networks." *Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (EMNLP-IJCNLP)*. 2019.

Steenbergen, Marco R., et al. "Measuring political deliberation: A discourse quality index." *Comparative European Politics* 1.1 (2003): 21-48.

Wang, Angelina, Jamie Morgenstern, and John P. Dickerson. "Large language models that replace human participants can harmfully misportray and flatten identity groups." *Nature Machine Intelligence* 7.3 (2025): 400-411.

Zhuang, Kaiyao, Yuheng Wang, and Weiyu Zhang. "Can AI Deliberate? Evaluating Deliberative Quality and Stance Flow in Multi-Agent LLMs." *Proceedings of the Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems*. 2026.